



UNIFIED ERP — ANY INDUSTRY

Procurement

RFQ → Purchase Order → Goods Receipt → 3-Way Match → Vendor Scorecard

Module 1 · Supply Chain & Procurement — App 1 of 6

An industry-neutral procure-to-pay module. The same engine runs a textile mill, a medical distributor, a manufacturer, or a services firm — you change only the vendor and item master.

Single-file · offline

14/14 self-tests

Wired to Stock · Finance · Planning

What this is & what's inside

Procurement is the **buy side** of the ERP: it turns a need into a quote, a quote into a purchase order, a delivery into accepted stock, and a supplier bill into a **safe, matched payment** — while scoring every vendor on real performance.

It is one self-contained HTML file. It opens by double-click, runs fully **offline**, saves your data in the browser, and checks itself with **14 self-tests** on every launch. This document tours all nine screens with live screenshots, then documents the exact formulas, the wiring to the rest of the ERP, and the tests.

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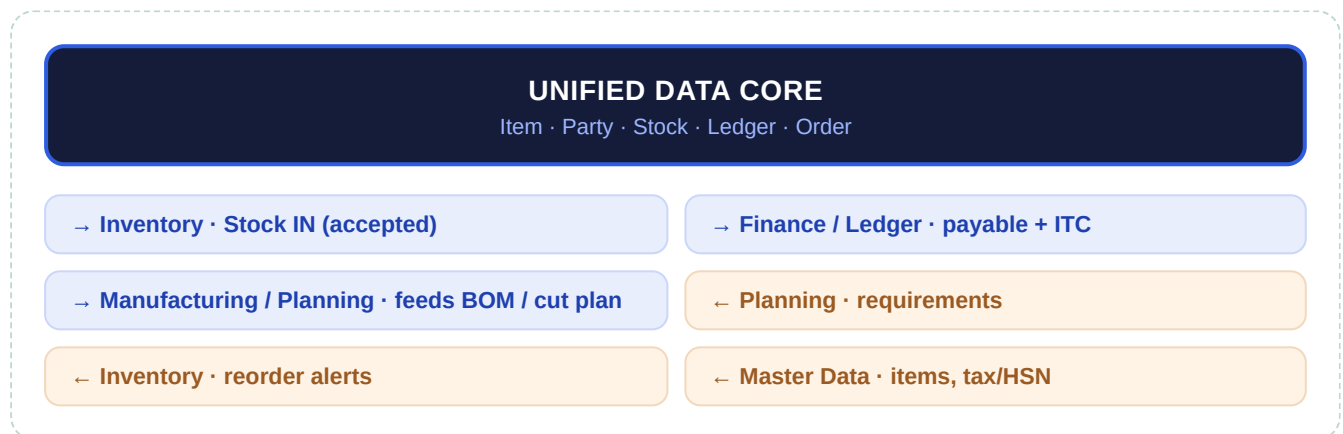
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Where Procurement sits

Every app in Medhava sits on **one shared Data Core + event bus**: Item/SKU, Party (vendor), Stock, Ledger/Voucher, Order. Procurement **writes** to Stock (accepted goods), the Ledger (payable + tax credit) and Master Data (preferred vendors), and **reads** requirements from Manufacturing/Planning and reorder alerts from Inventory.



The procure-to-pay pipeline. The scorecard runs across the whole history in parallel.



Data model — the entities

Procurement owns five documents; everything numeric is **computed**, never re-keyed.

ENTITY	KEY FIELDS
Vendor	id · name · gstin · category · terms
Item	code · name · uom · stdRate (benchmark)
Purchase Order	id · vendor · date · expected · status (draft/approved/partial/received/closed) · lines[item · qty · rate · tax%]
GRN (Goods Receipt)	id · po · date · onTime · lines[item · ordered · received · accepted · rejected]
Invoice / Bill	id · po · vendor · date · lines[item · qty · rate · tax%]
RFQ	id · item · qty · status · awarded · quotes[vendor · rate · lead days]

Invariants enforced by the engine: **accepted + rejected = received**, **received ≤ ordered**, and Input Tax Credit accrues on **accepted value only**.

Dashboard

The one-screen state of the buy side. Every figure is computed live from your documents — open commitments, receipts still due, blocked bills, and the tax credit you can claim.

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Acme Corp

FY 2026-27

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Command - Dashboard

Procurement — live

Open commitments, receipts pending, and 3-way exceptions — all computed from your documents.

OPEN POS

2

awaiting full receipt

OPEN PO VALUE

₹35,700.00

committed spend

PENDING GRN

2

to receive

3-WAY EXCEPTIONS

2

blocked bills

ITC CLAIMABLE

₹2,559.00

on accepted value

Recent purchase orders

PO	VENDOR	STATUS	GROSS
PO-1004	Delta Trading	partial	7,350.00
PO-1003	Gamma Materials Co	approved	28,350.00
PO-1002	Beta Components Ltd	received	8,400.00
PO-1001	Alpha Industrial Supplies	received	29,400.00

Raise a quick PO

VENDOR

Alpha Industrial Supplies

MATERIAL

Primary raw material

QTY

50

RATE ₹

100

GST %

5

Create PO

Open POs, open value, pending receipts, 3-way exceptions and claimable ITC — plus a quick-PO form.

- **Open PO value** sums the gross of every not-yet-received order.
- **3-way exceptions** is your risk number — bills that must not be paid yet.
- **Raise a quick PO** creates an approved order that immediately appears under Goods Receipt.

RFQ & quote comparison

Invite quotes from several vendors, compare landed rate and lead time side by side, and award the best in one click — the award becomes an approved purchase order automatically.

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RFQ & quote comparison

Invite quotes, compare landed rate, and award to the best vendor — that award becomes a PO.

RFQ RFQ-01 — Finishing consumable × 20 kg

open

VENDOR	QUOTED RATE	LEAD DAYS	LINE VALUE
Epsilon Enterprises	220.00	7	4,400.00
Alpha Industrial Supplies	240.00	5	4,800.00
Gamma Materials Co	210.00	9	4,200.00

Award lowest → create PO

lowest

Competing quotes with the lowest tagged; "Award lowest → create PO".

- ▶ The engine tags the **lowest rate** and, on award, spawns a PO to that vendor at that price.
- ▶ In the demo, the finishing-consumable RFQ awards to **Gamma Materials Co @ ₹210**.

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Purchase Orders

Each PO shows its lines with per-line net and tax, then rolls up to Net, GST and Gross. Draft orders are approved here; approved orders are received under Goods Receipt.

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Buying - Purchase Orders

Purchase orders

Each PO with lines, tax and gross. Approve draft POs; receive them under Goods Receipt.

PO-1004 — Delta Trading

partial

MATERIAL	QTY	RATE	GST%	NET	TAX
Lining / substrate	200	35.00	5	7,000.00	350.00
Net					₹7,000.00
GST					₹350.00
Gross					₹7,350.00
Expected					2026-07-08

PO-1003 — Gamma Materials Co

approved

MATERIAL	QTY	RATE	GST%	NET	TAX
Secondary raw material	300	90.00	5	27,000.00	1,350.00
Net					₹27,000.00
GST					₹1,350.00
Gross					₹28,350.00
Expected					2026-07-12

PO-1002 — Beta Components Ltd

received

MATERIAL	QTY	RATE	GST%	NET	TAX
Precision component	50	150.00	12	7,500.00	900.00
Net					₹7,500.00
GST					₹900.00
Gross					₹8,400.00
Expected					2026-07-04

PO-1001 — Alpha Industrial Supplies

received

MATERIAL	QTY	RATE	GST%	NET	TAX
Primary raw material	100	280.00	5	28,000.00	1,400.00
Net					₹28,000.00
GST					₹1,400.00
Gross					₹29,400.00
Expected					2026-07-03

Line detail with Net/Tax, and PO Net → GST → Gross → Expected date.

- **Net** = $\sum(\text{qty} \times \text{rate})$; **Tax** = $\sum(\text{qty} \times \text{rate} \times \text{GST}\%)$; **Gross** = Net + Tax.
- Status flows draft → approved → partial → received as goods arrive.

Goods Receipt (GRN)

The moment quality matters. Against each PO you record **received**, **accepted** and **rejected**. Only the **accepted** quantity posts to Stock and earns tax credit; rejects raise a debit note.

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Buying - Goods Receipt

Goods receipt (GRN)

Received, accepted and rejected against each PO. Accepted quantity is what posts to Stock and earns ITC.

GRN-503 — PO-1004 (Delta Trading)

on-time

MATERIAL	ORDERED	RECEIVED	ACCEPTED	REJECTED
Lining / substrate	200	180	180	0
→ Stock IN (accepted)				180

GRN-502 — PO-1002 (Beta Components Ltd)

late

MATERIAL	ORDERED	RECEIVED	ACCEPTED	REJECTED
Precision component	50	50	50	0
→ Stock IN (accepted)				50

GRN-501 — PO-1001 (Alpha Industrial Supplies)

on-time

MATERIAL	ORDERED	RECEIVED	ACCEPTED	REJECTED
Primary raw material	100	100	96	4
→ Stock IN (accepted)				96
→ Rejected (debit note)				4

Receive an approved PO

PO

PO-1003 · Gamma Materials Co

RECEIVED QTY

0

REJECTED QTY

0

Post GRN

Ordered/received/accepted/rejected, with "→ Stock IN (accepted)" and rejects flagged.

- ▶ Receiving clamps to ordered; a partial receipt sets the PO to **partial**.
- ▶ GRN-501 accepts 96 of 100 — so Stock rises by 96, not 100, and the 4 rejects become a debit note.

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3-Way Match — the control

The heart of procurement. A supplier bill only passes for payment when **PO, GRN and Invoice agree** on price and quantity. Any mismatch is held with a plain-English reason.

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Control - 3-Way Match

Three-way match

PO → GRN → Invoice. A bill only passes when price, received and ordered quantities all agree.

BILLS CHECKED

2

with invoice

MATCHED

0

ready to pay

EXCEPTIONS

2

blocked

BILL-9001 — Alpha Industrial Supplies

exception

DOC	QTY	RATE
PO PO-1001	100	280.00
GRN GRN-501	96 acc	—
Invoice BILL-9001	100	280.00

Held: billed 100 > accepted 96

BILL-9002 — Beta Components Ltd

exception

DOC	QTY	RATE
PO PO-1002	50	150.00
GRN GRN-502	50 acc	—
Invoice BILL-9002	50	165.00

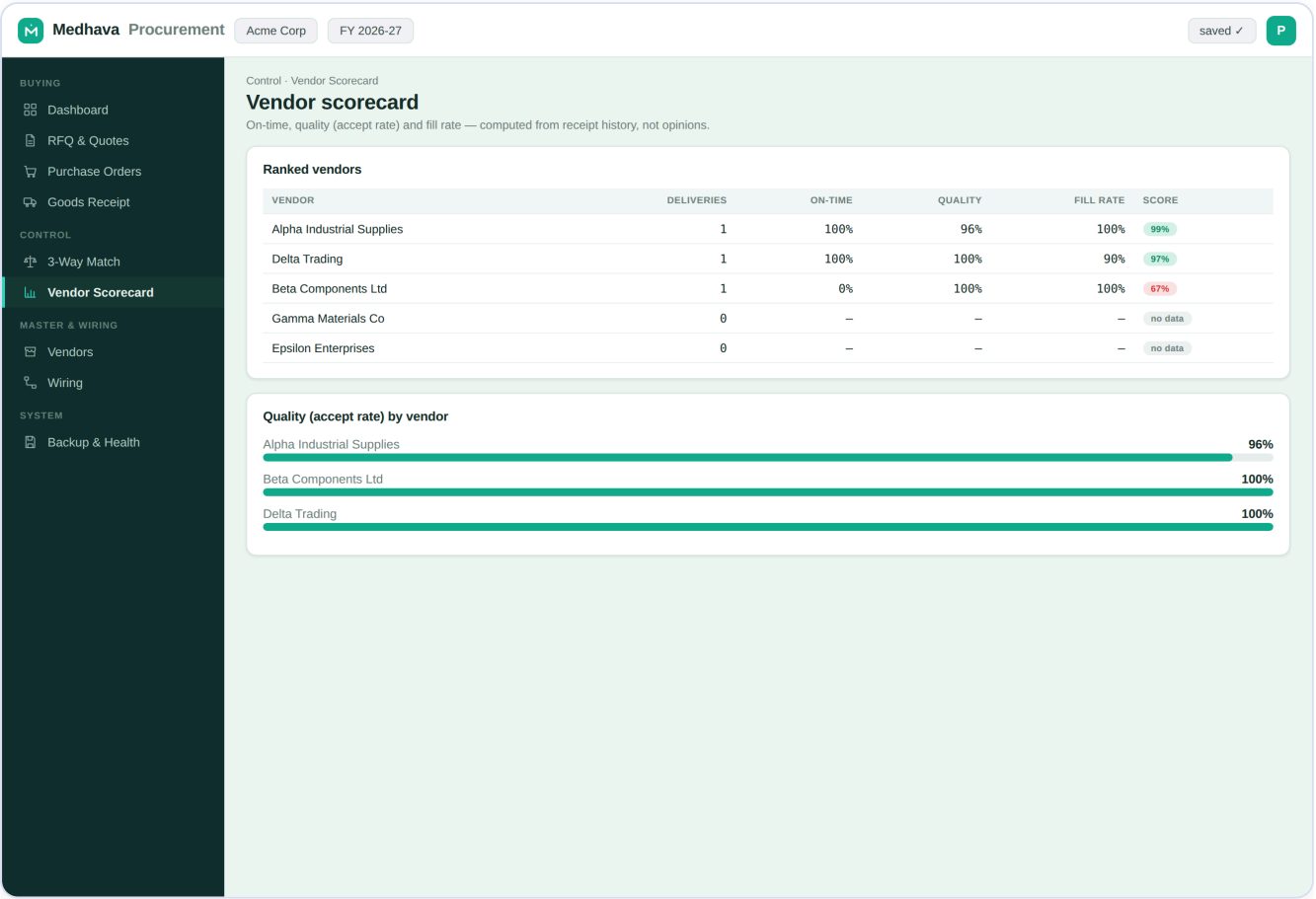
Held: price 150.00 → 165.00

Two demo bills, both correctly held: one over-bills rejects, one hikes the price.

- ▶ **BILL-9001** — held: *billed 100 > accepted 96* (paying for rejected goods).
- ▶ **BILL-9002** — held: *price 150 → 165* (rate above the PO beyond 0.5% tolerance).
- ▶ This single screen is what stops silent margin leakage on the buy side.

Vendor Scorecard

Vendors ranked on facts, not opinions — on-time %, quality (accept rate) and fill rate, all computed from receipt history, rolled into one score.



Ranked vendors with on-time / quality / fill and a colour-coded score, plus quality bars.

- ▶ **Alpha Industrial Supplies:** 96% quality (96/100), 100% on-time.
- ▶ A late delivery drops on-time to 0% for that vendor; a short delivery shows as a <100% fill rate.

Vendors & item master

The single source every PO, GRN and bill draws from — vendors with GSTIN, category and terms, and items with unit of measure and a benchmark rate for price checks.

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Master - Vendors

Vendor master

GSTIN, category and payment terms — the single source every PO and bill draws from.

Vendors

ID	VENDOR	GSTIN	CATEGORY	TERMS
V1	Alpha Industrial Supplies	27AAAAA0001A1Z1	Raw material	30 days
V2	Beta Components Ltd	29BBBBB0002B1Z2	Components	15 days
V3	Gamma Materials Co	24GGGGG0003C1Z3	Raw material	45 days
V4	Delta Trading	07DDDDD0004D1Z4	Consumables	30 days
V5	Epsilon Enterprises	19EEEEEE0005E1Z5	Packaging	COD

Item master (materials)

CODE	NAME	UOM	STD RATE
ITM-01	Primary raw material	kg	280.00
ITM-02	Secondary raw material	kg	90.00
ITM-03	Precision component	pc	150.00
ITM-04	Lining / substrate	m	35.00
ITM-05	Finishing consumable	kg	210.00

Vendor master and item master — the shared reference data.

- ▶ Changing a vendor's terms or an item's std rate flows to every future document.
- ▶ The std rate is the yardstick the 3-way match and repricing use.

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Wiring — one action cascades

Procurement never stands alone. This screen makes the integration literal: every event writes to the shared Data Core, so a single Goods Receipt updates Stock, Finance, Quality and the Scorecard at once.

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Wiring - Integration

How Procurement wires into the rest of the ERP

Procurement never stands alone. Each event here writes to the shared Data Core, so one action cascades everywhere.

Shared Data Core: Item/SKU · Party (vendor) · Stock · Ledger/Voucher · Order — every module reads and writes these.

Outbound data flows (this app → others)

TRIGGER (HERE)	FLows TO	WHAT MOVES
GRN accepted qty	Inventory / Stock	Stock IN at receiving location (single source of truth)
Supplier invoice (matched)	Finance / Ledger	Vendor payable + Input Tax Credit on accepted value
GRN rejected qty	Quality + Finance	Debit note to vendor + quality flag on scorecard
PO approved	Budget / Commitments	Committed spend against cost centre / budget
RFQ awarded	Master Data (Party)	Preferred-vendor + agreed price for the item
Vendor scorecard	Master Data	On-time / quality rating drives future sourcing

Live example — GRN-501 cascade

1 Goods Receipt GRN-501 — 96 accepted, 4 rejected.

2 → **Inventory**: Stock IN +96 Primary raw material at the receiving location.

3 → **Finance / Ledger**: payable to Alpha Industrial Supplies + ITC on accepted value (₹1,344.00).

4 → **Quality**: 4 rejected raise a debit note to the vendor + a quality flag on the scorecard.

5 → **Vendor Scorecard**: on-time + accept-rate recomputed from this receipt.

Inbound (others → Procurement)

FROM	WHAT
Manufacturing / Planning	Reorder & material requirements raise purchase requisitions
Inventory	Below-reorder alerts trigger a PO suggestion
Master Data	Item master, vendor master, tax/HSN rates
Automation	"If stock < reorder then draft PO to preferred vendor"

Outbound/inbound data flows and the live GRN-501 cascade with ITC ₹1,344.00.

- ▶ GRN-501 → Stock IN +96 → Finance / Ledger payable + ITC ₹1,344.00 → debit note on 4 rejects → scorecard recompute.
- ▶ Inbound: Manufacturing / Planning requirements, Inventory reorder alerts, Master Data, Automation.

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Formulas & business rules

```
lineNet(l) = r2(qty × rate)
lineTax(l) = r2(qty × rate × tax/100)
poNet      = r2(Σ lineNet)      poTax = r2(Σ lineTax)      poGross = poNet + poTax

GRN        accepted + rejected = received      received ≤ ordered
ITC        = r2( Σ_grn Σ_line ( accepted × P0.rate × P0.tax/100 ) )      // accepted value only

3-WAY MATCH (tolerance TOL = 0.5%) → exception if any:
|invoice.rate - P0.rate| / P0.rate > TOL      → "price X→Y"
invoice.qty > GRN.accepted                    → "billed N > accepted M"
invoice.qty > P0.qty                          → "billed N > ordered M"

SCORECARD  onTime% = onTimeGRNs / totalGRNs ×100
           quality%= accepted / received ×100
           fill%   = received / ordered ×100
           score   = mean(available metrics)      badge ≥90 grn · 70–89 amb · <70 red

RFQ AWARD  winner = min(rate) → creates approved P0 at winning rate
```

Everything above is **derived** at read time. The database stores documents; the numbers are always recomputed, so they can never drift out of sync.

Self-tests — all 14 pass

The app runs these on every launch and shows the result on **Backup & Health**. They pin the exact numbers this document quotes, so "it works" is verifiable, not asserted.

#	CHECK	EXPECTED	RESULT
1	PO net = qty × rate	poNet (P0-1001) = 28,000	✓ pass
2	PO tax = qty × rate × gst%	poTax = 1,400	✓ pass
3	PO gross = net + tax	29,400	✓ pass
4	GRN accepted + rejected = received	96 + 4 = 100	✓ pass
5	GRN received ≤ ordered	100 ≤ 100	✓ pass
6	ITC on accepted value only	2,559.00	✓ pass
7	3-way flags the price mismatch	BILL-9002 (150→165)	✓ pass
8	3-way flags over-billing vs accepted	BILL-9001 (100 > 96)	✓ pass
9	exactly 2 bills, both exceptions	matched 0 / exceptions 2	✓ pass
10	V1 quality = 96%	96 / 100 received	✓ pass
11	V1 on-time = 100%	1 / 1 delivery	✓ pass
12	V2 on-time = 0%	late delivery	✓ pass
13	V4 fill rate = 90%	180 / 200 ordered	✓ pass
14	RFQ award picks lowest quote	winner @ ₹210	✓ pass

Verified twice: engine tests in Node, then all nine screens in a real headless browser with zero console errors.

Data in / out & hosted API

The single-file app is the **local edition**. The hosted, multi-tenant edition runs the identical engine against a backend so Stock and Ledger update across modules for real.

BOUNDARY	LOCAL (THIS FILE)	HOSTED EDITION
Storage	Browser localStorage + JSON backup	Postgres with row-level security per company
Stock update	Shown as " → Stock IN"	Event bus posts to the real Stock ledger
Books	Shown as payable + ITC	Double-entry voucher in Finance / Ledger
Vendor connect	Manual entry	Vendor portal + email/RFQ ingest
Auth to 3rd parties	—	Revocable scoped API keys in an encrypted vault — never passwords

API surface (hosted): /rfq · /po · /grn · /invoice · /match · /vendor/scorecard — each emits an event other modules subscribe to.

Honest limits & roadmap

What this build is — and isn't — so there are no surprises.

- ▶ **Local-first, single browser.** Cross-module posting is real in the hosted edition, illustrated [here](#).
- ▶ **One line per document** in the demo seed; the engine already supports multi-line POs/GRNs/bills.
- ▶ **Not yet in this file:** multi-currency, landed-cost apportionment, batch/expiry, vendor-portal quoting — all hosted-tier.
- ▶ **Security:** real connections use scoped, revocable keys in an encrypted vault; account passwords are never used or stored.

Roadmap for this module



Module 1 · Supply Chain & Procurement — the other five apps follow, same depth.

How to run & accept

1. **Open** `procurement_ERP.html` by double-clicking — any modern browser, offline.
2. **Backup & Health** → confirm **14/14 self-tests pass**.
3. **3-Way Match** → BILL-9001 & BILL-9002 are held with reasons.
4. **Goods Receipt** → each GRN shows " → Stock IN (accepted)".
5. **Vendor Scorecard** → Alpha Industrial Supplies 96% quality / 100% on-time.
6. **RFQ** → "Award lowest → create PO" → Gamma Materials Co @ ₹210.
7. **Wiring** → the GRN-501 cascade (ITC ₹1,344.00) proves the numbers flow onward.

Accepted when: it opens offline, all 14 tests are green, no console errors, and the exported files match your own records.

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